

OptionsAgent

A paper-trading bot that sells credit spreads. Claude proposes. Code decides.

PAPER ONLY

CREDIT SPREADS

RUNS ON THIS MAC

LLM NEVER PICKS STRIKES

OptionsAgent trades one strategy on a **\$100,000 Alpaca paper account**: defined-risk credit spreads on 13 liquid names under \$50. Once a day it asks Claude for trade ideas, but Claude only gets to say **which ticker, which direction, and how confident**. Every strike, expiration, size, and exit is chosen by fixed Python rules called rails. If an idea fails any rail, it dies. Most days, nothing trades, and that is by design.

STEP 1

Cron wakes it

Weekdays **10:15 to 10:27 ET**, once a day. A lock file blocks double runs. If the broker clock check fails or the market is shut, it skips the day.

STEP 2

Snapshot

Archives the full option chain (quotes, IV, greeks) for all **13 watchlist names**, then builds a market context bundle per name.

STEP 3

Claude proposes

The local Claude CLI ([sonnet](#), no tools, no web) reads the bundle and proposes **ticker, direction, conviction 0 to 1, thesis**. Strikes and size are banned from its output. CLI failure means zero trades.

STEP 4

Rails veto or size

Deterministic gate: conviction floor **0.60**, max **6** open positions, phase allowlist. Size scales from 30% to 100% of available options buying power.

STEP 5

Code picks the contract

Short leg **15 to 30 delta**, width capped at **\$2, 30 to 45 DTE**. Bullish sells a put spread, bearish sells a call spread. One combo limit order at the net credit.

STEP 6

The winner profile

The spread must also match one of three replayed winners: **CCL bullish put, SOFI bullish put, F bearish call**. Everything else is vetoed. Quiet days are normal.

MEANWHILE, EVERY 20 MIN

A separate exit sweep runs with **no LLM at all**: take profit at **50% of the credit**, stop out at **2x credit** (only after 10:00 ET and only after two consecutive sweeps agree, so one wide quote cannot force a liquidation), force close at **21 DTE**, and check ex-dividend dates before any short call sits through one.

THE NUMBERS THAT GOVERN IT

Account	\$100,000 paper (re-funded Aug 28)
Conviction floor	0.60, below = no trade
Sizing	30% to 100% of free BP
Max concurrent	6 positions
Spread width	≤ \$2.00, max loss ≈ \$200
Entry DTE	30 to 45
Profit target	50% of credit
Stop	2x credit, confirmed twice
Force close	21 DTE

THE WATCHLIST (13 LIQUID NAMES UNDER \$50)

F T PFE VZ CMCSA KVUE SOFI CCL AAL WBD MARA EWZ KWEB

WHAT IT WILL NEVER DO

- ✗ Let model output pick a strike, delta, or expiration
- ✗ Loosen a rail by editing config (env vars may only tighten)
- ✗ Place a live-money order (the broker client refuses non-paper)
- ✗ Buy options after an extended move (the anti-chase rule)

A REAL DAY (AUG 28, 2026)

Claude proposed **8 spreads** across T, CMCSA, WBD, VZ, PFE, MARA, CCL, and AAL. The rails executed **0**. Three found no contract meeting the delta, width, and credit rules. Five failed the winner profile. Meanwhile the exit sweep closed July's SOFI spread for **+\$26.58**. Vetoing everything is the system working, not failing.

FAQ

Everything below is verified from the current code and the live paper account, not from the older docs.

What strategy is running right now, exactly?

Vertical credit spreads only. Bullish thesis sells a put spread, bearish sells a call spread, never neutral. Short leg between 15 and 30 delta, long leg further out on the same expiry, width capped at \$2.00, entry between 30 and 45 DTE. The spread is submitted as a single MLEG combo limit order priced at the net credit, order IDs prefixed oa–.

How does position sizing work?

Position budget = conviction-scaled share of available options buying power. At the 0.60 floor you get 30% of what is free, scaling linearly to 100% at 0.85 and above. $\text{Contracts} = \text{budget} \div (\text{width} \times 100)$. If the budget buys less than one contract, the trade is skipped. Each fill inside a cycle shrinks the base the next proposal is sized against.

100% of buying power on one idea sounds aggressive.

It is deliberate operator policy: "use all the cash, not necessarily all at once." The practical brakes are the \$2 width cap (max loss $\approx$ \$200 per spread, 0.2% of the account), the 6 position cap, the broker's own buying power rejection, and an optional tighten-only kill switch (OA_MAX_POSITION_USD) that can clamp any single position.

Where does market data come from?

Option chains and quotes from a read-only [Public.com](#) sidecar; execution and account state at Alpaca paper; equity snapshots from the broker. Every entry cycle also archives the full chain (quotes, IV, greeks) to disk so a backtester can be built later from real captured data.

What is the winner profile gate and why does it veto so much?

A replay of every archived fill from July 8 to July 31 found three profiles that actually made money: CCL bullish puts (credit $\geq$ \$0.29, width $\geq$ \$1.50), SOFI bullish puts (credit $\geq$ \$0.23, width $\geq$ \$1.00), and F bearish calls (credit $\geq$ \$0.06, width $\leq$ \$0.50). Those three are hard-coded. Anything else fails closed. It is openly an in-sample overfit, held frozen on purpose so the account collects clean out-of-sample evidence on whether the profile means anything.

Walk me through the exits.

A no-LLM sweep every 20 minutes during market hours. Priority: dividend risk first (a short call sitting through an ex-date where the dividend exceeds the same-strike put gets closed), then the stop, then DTE, then profit. Profit target is 50% of the credit received. The stop is 2x credit but only counts from 10:00 ET and needs two consecutive sweeps to confirm, so a single wide opening quote cannot trigger a forced liquidation. Everything force-closes at 21 DTE. There is no end-of-day flatten for spreads because nothing here is 0DTE.

What about assignment and overnight gap risk?

Verticals are defined risk: worst case is width minus credit, no more, regardless of assignment mechanics. The 2x credit stop is a loss control, not a guarantee. A gap through the short strike overnight still costs the full width, roughly \$200 per spread at the cap. Six concurrent positions put the theoretical worst day near \$1,200, just over 1% of the \$100,000 account.

Is there a daily loss limit for the seller?

No dedicated one. The brakes are structural: \$2 max width, 6 positions, one short entry window per day (10:15 to 10:27 ET), and the winner profile that rejects most ideas before they exist. If that answer makes you uncomfortable, that is a fair reaction; it is a paper account built to learn the distribution.

What does the LLM see, and can it be manipulated by the data?

It sees a JSON bundle: the phase, the allowed strategies, and per-ticker market context (price levels, volume, recent movement). The system prompt states the context is data, never instructions. It runs with tools disabled, no session persistence, no web access, and its output is validated against a strict schema before anything downstream sees it.

How do I know what it did on any given day?

Three places. The dashboard on the following pages (live at 127.0.0.1:8765, read-only, it cannot submit orders). The decision log at `data/decisions.jsonl`, which records every proposal, every veto reason, and every order. And `data/structures.jsonl`, the append-only record of what opened and closed. Today's log shows 8 proposals and 8 vetoes with reasons.

What stops a bug from doing something catastrophic?

Layers. Rails are hard-coded in Python so no config edit or model output can loosen them; env overrides may only tighten. The broker client refuses non-paper endpoints. Entry is one locked window a day. The exit sweep reconciles the registry against live broker positions and alerts on anything that vanished (assignment, expiry, manual close) instead of silently dropping it.

OptionsAgent

Multi-day credit spread seller

PAPER · ACTIVE

Overview

Positions

Trade history

Research

Risk rails

System

Friday, August 282026

Paper account

11:49 AM ET

Today P/L

↗

+\$0

Realized, market open

Open spreads

▮

0 of 6

Total open positions

Buying power

▮

\$100,000

Available to deploy

Profile status

◇

3 winner rules

Active

Equity curve

30D

Day	Equity
30D	\$0
29D	-\$100
28D	-\$1,789
27D	-\$1,800
26D	-\$1,850
25D	-\$1,880
24D	-\$1,900
23D	-\$1,920
22D	-\$1,950
21D	-\$1,980
20D	-\$1,990
19D	-\$1,990
18D	-\$1,980
17D	-\$1,970
16D	-\$1,960
15D	-\$1,950
Today	+\$0

Daily P/L, realized

30D

Day	P/L
1	-\$1,789
2	-\$100
3	-\$100
4	-\$100
5	-\$100
6	-\$100
7	+\$100
8	+\$100
9	+\$100
10	+\$100
11	+\$100
12	+\$100
13	+\$100
14	+\$100
15	+\$100
16	+\$100
17	+\$100
18	+\$100
19	+\$100
20	+\$100
21	+\$100
22	+\$100
23	+\$100
24	+\$100
25	+\$100
26	+\$100
27	+\$100
28	+\$100
29	+\$100
30	+\$100

Last 30 days8 days

Credit spread history

View all →

Research replay

⚡

Live local dashboard · 127.0.0.1:8765

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Multi-day credit spread seller

PAPER · ACTIVE

- Overview
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OptionsAgent v1.0.0

Friday, August 28

2026

Paper account

11:49 AM ET

Positions

Live paper positions from the broker snapshot.

Overview

SYMBOL	STRATEGY	QTY	MARKET VALUE	COST BASIS	STATUS
No broker positions reported.					

Read-only dashboard · Account data is cached and never used to submit orders.



Multi-day credit spread seller

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Overview

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Paper account

11:49 AM ET

Trade history

Closed structures from the append-only registry. Unknown P/L stays unknown.

Overview

SYMBOL	TYPE	WIDTH	CREDIT	P/L	EXIT
SOFI	Credit Spread	\$0.50	\$0.08	—	8/28/2026, 9:40 AM
F	Credit Spread	\$0.50	\$0.06	+\$45	7/31/2026, 1:20 PM
SOFI	Credit Spread	\$1.00	\$0.23	+\$40	7/31/2026, 9:40 AM
MARA	Credit Spread	\$1.00	\$0.23	-\$25	7/31/2026, 9:40 AM
SOFI	Credit Spread	\$0.50	\$0.11	+\$0	7/30/2026, 9:40 AM
CCL	Credit Spread	\$1.50	\$0.29	+\$45	7/28/2026, 11:40 AM
AAL	Credit Spread	\$1.00	\$0.19	-\$165	7/23/2026, 10:20 AM
AAL	Credit Spread	\$1.00	\$0.19	-\$57	7/16/2026, 10:20 AM
VZ	Credit Spread	\$1.00	\$0.17	-\$221	7/10/2026, 9:40 AM

The replay studies behind the rules: the scalp entry cutoff (12 of 12 late entries lost), the credit-spread winner mining, and the in-sample P/L (+\$345 on 12 retained trades) the profile gate is built on.

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2026

Paper account 11:49 AM ET

Research replay

← Overview

Conditional realized replay; not a full backtest or durable edge claim.

Replay summary	
Entry days	5
Archived records	10
Realized P/L	-\$644
Filtered profile replay	+\$130
Unknown P/L closes	1

Winner rules	
CCL bullish · width ≥ \$1.50 · credit ≥ \$0.29	Active
SOFI bullish · width ≥ \$1.00 · credit ≥ \$0.23	Active



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Friday, August 28

2026

Paper account

11:49 AM ET

Risk rails

Deterministic gates remain in code and the bot is paper-only.

← Overview

Deterministic hard rails	
Rollout phase	credit_spreads_only
Conviction floor	0.6
Maximum concurrent positions	6
Mandatory close	21 DTE
Execution mode	PAPER ONLY

Read-only dashboard · Account data is cached and never used to submit orders.



Multi-day credit spread seller

PAPER · ACTIVE

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OptionsAgent v1.0.0

Friday, August 28
2026

Paper account 11:49 AM ET

System

Runtime health and market-data provider status.

Overview

Runtime	
Trading gate	Enabled
Broker mode	Paper
Market-data provider	public
Snapshot	8/28/2026, 11:48:49 AM ET
Refresh failures	0

Read-only dashboard · Account data is cached and never used to submit orders.